CPSC4140
Fall 2026
10/05/2026

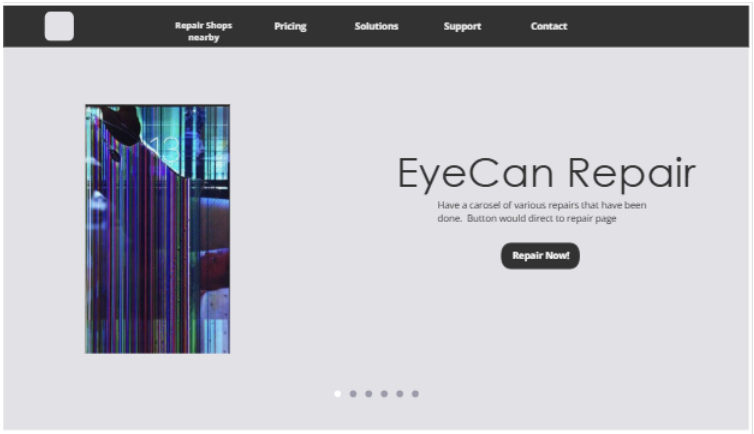
Milestone II: Drastically Different Designs

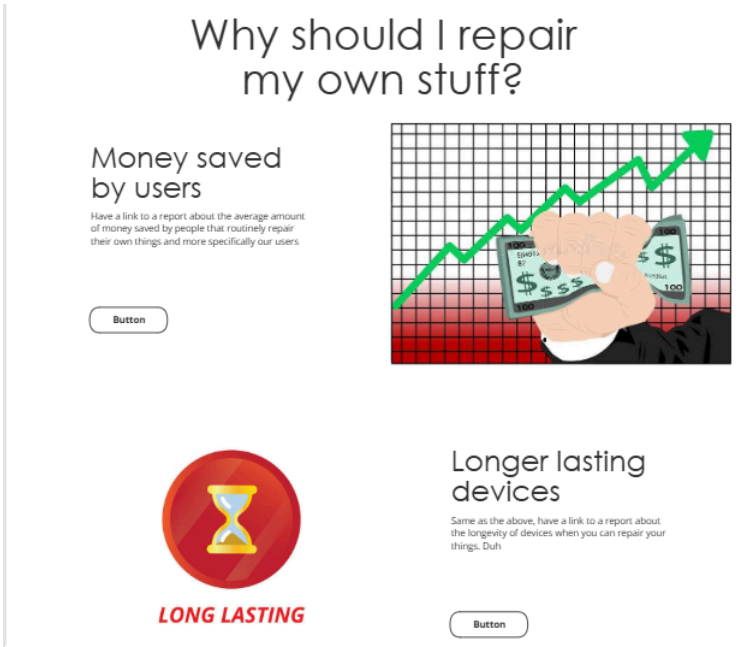
The Two-Weekers
Team 05
Rayan Ahmed, Miller Hendricks, Ford Scott, Shaun Whitt

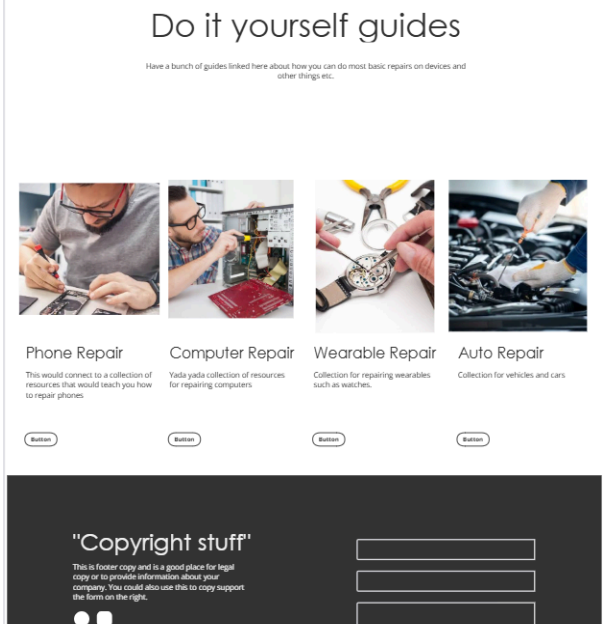
Idea 1: Repair Workbench Web Application

<u>Mockups</u>

General Idea of the Main page







Detailed Narrative

This idea would exist solely on the internet for potential users to use either by intentionally looking for the site or when they are conducting general repair research. The site would be designed to both inform inexperienced users about the benefits of repairing their own technology and connect them to resources teaching them how to do it. Other resources we would provide would include a map showing them the nearest repair shops in their area for what they are looking to fix, analyzing what exactly the user is trying to fix and showing what the cost of said repair would be compared to a replacement, and a more specific page about how to diagnose what is wrong with your technology. These pages would have some degree of overlap as, for example, when a user is given their estimate of how much it would cost to repair they would then have the option of viewing what shops are nearby for them to do the repair at.

Design Checklist

What makes this design fundamentally different from the other two?

- Must be consistently available as a web application
- Is geared towards the widest array of an audience: including both those curious about repairing their tech and those who are experienced.
- Has the most free form version of an interaction model, as users are not restricted to what they are allowed to do at any given time

What assumptions does this design make about the user and their situation?

- The user must be adept at navigating websites
- The user must be at least somewhat interested in repairs
- The user is currently having an issue with their technology
- The user is not on a time crunch

How does the interaction flow differ from the other designs?

In this idea the task flow is entirely up to the user as they are not initially prompted to do anything and are free to navigate the website to complete whatever task they would like, provided it is one we provide solutions for. Our other two ideas are geared towards solving specific issues their devices may be dealing with, rather than a general exploration of device repair. Of course this idea does also offer solutions towards the issue facing a specific device but that is not what it is designed solely to do.

How is this design informed by findings from Milestone 1?

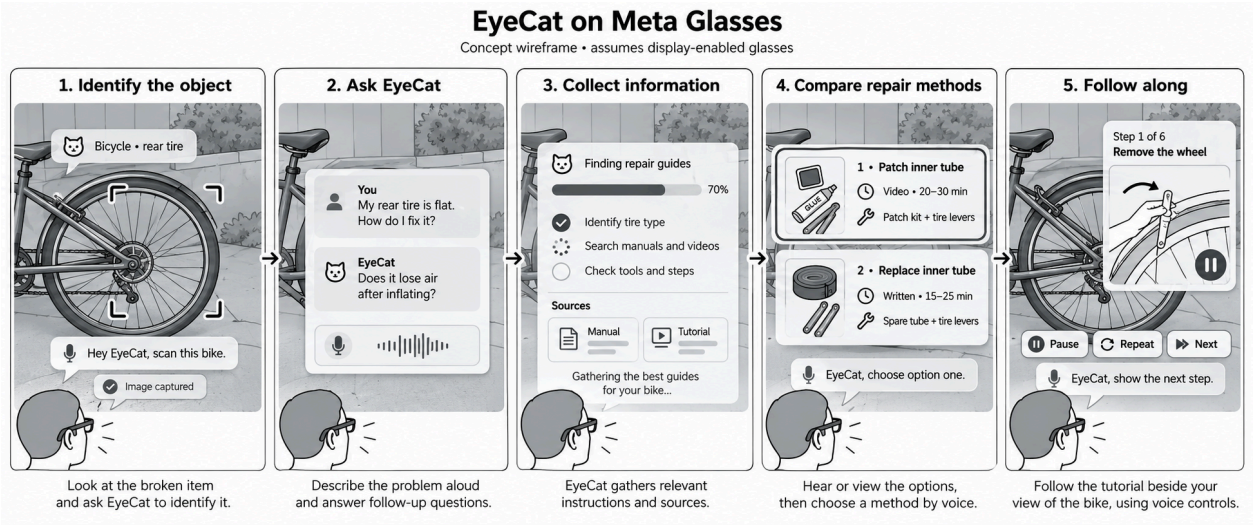
The main issue our peers warned us about from Studio 1 was that our problem space was fairly oversaturated so we needed to make sure that whatever ideas we came up with needed to be unique and not be just another tired app that has been made by 100s of other people. This is largely reflective in all of our ideas since the self-repair problem is fairly untouched, in comparison to our larger problem space of becoming more frugal with your money.

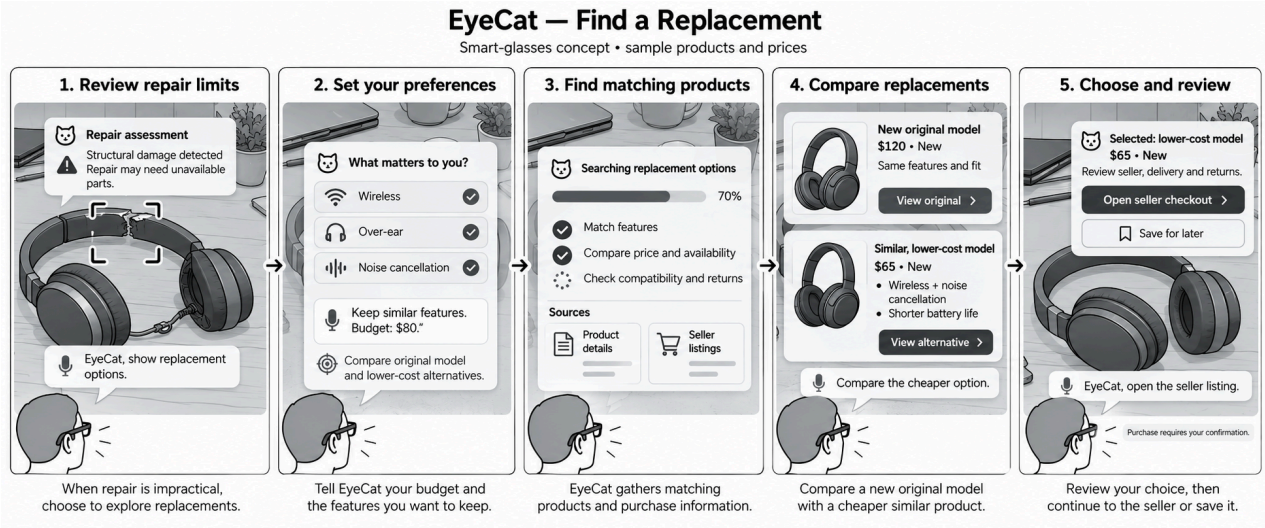
Strengths and Limitations

The main strength of this idea is that it is the most accessible version to the widest audience. It also holds the greatest variety of solutions to tasks/problems a user might have. The next main strength of this idea is that it would also be the best solution for those with impairments as we would be able to implement the largest amount of accessibility options as compared to the other two options. The main weakness of this idea is that since it is built for a wide audience, it inevitably leads to the depth of the solutions provided suffering. Another weakness to this idea is that it would be one of the more expensive options as if it were fully implemented the idea would require constant upkeep charges and since we would not be charging for anything, those costs would add up.

Idea 2: Smart Glasses with Voice Repair Assistant

Mockups





Detailed Narrative

People are constantly on the move, and finding help shouldn't slow them down. EyeCat is a portable, easy-to-use AI assistant that helps people diagnose, repair, and find replacements for everyday products wherever they are. To get started, users would connect their compatible smart glasses, enable camera and microphone access, and follow a short setup check. First-time users could take an optional tutorial that demonstrates how to capture an object, ask questions, and control instructions by voice, while experienced users could skip directly to asking for help.

Instead of spending hours searching through forums, manuals, and videos, users can show EyeCat the item they're trying to fix. Using visual input, it can help identify the product, recognize visible damage, and suggest practical next steps. For example, a student who notices a flat bicycle tire during their commute could stop somewhere safe, look at the tire through their glasses, and say, "EyeCat, my tire is flat. Can you help me fix it?" EyeCat would use the camera view and ask follow-up questions, such as whether the tire loses air after being inflated. If it needs a clearer view or a product detail, it could ask the user to move closer or show a label.

Once it understands the problem, EyeCat searches for relevant repair guides, tutorials, and product information, then organizes the most useful options in one place. It can tailor suggestions to the user's skill level and available tools, explain what materials may be needed, and flag when a repair may require professional help. In the bicycle example, EyeCat could present options for patching or replacing the inner tube, with the tools, estimated time, and difficulty of each method. The user could say, "I've never done this before," to receive more detailed explanations, or ask for a shorter guide if they already have repair experience. Before starting, EyeCat would help them check whether they have the necessary supplies and a suitable place to work. This can make it easier to compare repair options before spending money on a replacement or service.

EyeCat can also guide users through a repair step by step. It can display a video or illustrated instructions alongside the user's view of the product, helping them follow each stage without repeatedly pausing to search for the next step. After choosing a method by saying, "Show me how to patch the tube," the user would see the first instruction through the glasses' display and hear an accompanying explanation. They could keep their hands free while saying "Pause," "Repeat that," or "Next step" to control the guide. Users can ask questions as they work, get an explanation of unfamiliar terms, and move forward at their own pace. For instance, asking "What is a tire lever?" could bring up a brief explanation and illustration before they continue. At the end of the guide, EyeCat would prompt the user to test the repair and report the result. If the tire still loses air, it could help them revisit the relevant steps or consider further assistance.

If a repair is too difficult, too expensive, or impossible, EyeCat can help users explore replacement options. A user examining headphones with a broken headband,

for example, might learn that the repair requires a part they cannot obtain. They could then say, “EyeCat, help me find a replacement,” without starting a separate search. EyeCat would ask about their budget, the features they want to keep, and whether they prefer the same model or an alternative. The user might respond, “Find wireless headphones with noise cancellation for under eighty dollars.”

Based on their budget and preferred features, it can compare a brand-new version of the original item with cheaper, similar products, highlighting differences in price, compatibility, and functionality. These options would appear as simple comparison cards, allowing the user to ask questions such as, “What would I lose by choosing the cheaper pair?” EyeCat could explain relevant tradeoffs, such as shorter battery life or fewer features, and refine the results if the user changes their preferences. Users can review seller information, delivery options, and return policies before choosing to continue to checkout or save an option for later. Once they select a product, EyeCat would present the details for review and direct them to the seller’s checkout, where they would confirm the purchase themselves. If they are not ready to buy, they could say, “Save this for later,” and return to the option when convenient.

By bringing visual assistance, repair resources, guided instructions, and replacement comparisons together, EyeCat aims to make dealing with broken products less frustrating and more accessible. The experience would give users a clear path from identifying a problem to attempting a repair or choosing a replacement, with explanations available whenever they need them. It can help people save time, avoid unnecessary costs, and feel more confident about deciding whether to repair or replace the products they own.

Design Checklist

What makes this design fundamentally different from the other two?

While our other two ideas share the goal of making repairs easier, EyeCat focuses on immediate access to personalized assistance wherever a problem occurs. For example, if your bike breaks down during your commute to school, you could put on your smart glasses and ask EyeCat to examine the bike, identify possible issues, and explain what tools or actions may be needed. A repair website provides guides and resources, but users still have to describe the problem, search for relevant instructions, and decide which guide fits their situation. EyeCat would use visual input and follow-up questions to help narrow down the problem and find suitable repair methods. A community repair kiosk offers a dedicated place to access assistance and equipment, but users must travel to that location. EyeCat brings guidance directly to the user, which could be especially helpful when a broken object is difficult to move. Its glasses-based interface would also allow users to keep their hands free while viewing instructions beside the object, asking questions aloud, and moving through each step at their own

pace. Although users would still need the appropriate tools and parts, EyeCat's combination of portability, visual recognition, and interactive guidance could reduce the time spent searching for answers and help people approach repairs with greater confidence.

What assumptions does this design make about the user and their situation?
EyeCat's design assumes that the user:

- Has access to compatible smart glasses and the internet. The glasses need sufficient battery power, and EyeCat needs connectivity to retrieve repair guides and replacement options.
- Can get started through a simple setup process. EyeCat should guide users through connecting their glasses, enabling camera and microphone access, and checking their internet connection. Clear prompts and a short practice interaction should make it easy to begin without technical knowledge.
- May be a complete beginner using EyeCat. First-time users may not know how to capture an object, ask for help, or navigate instructions by voice. An optional tutorial during setup should demonstrate these actions, including how to pause, repeat, and advance a step. Users should also be able to reopen the tutorial through a Help menu or by saying, "EyeCat, show me how to use this."
- May already be experienced with the software. Regular users should be able to skip the introductory tutorial and start by looking at an object and asking for assistance. Familiar voice commands and saved preferences could reduce the effort needed for repeat use.
- May be a beginner or an expert at repairs. Software experience does not necessarily mean repair experience. A beginner may need explanations of tools, terms, and each individual action, while an experienced user may prefer concise instructions or direct access to a source guide. EyeCat should let users adjust the level of detail and request a demonstration whenever a step is unfamiliar.
- Can show the problem clearly. The environment needs enough light, and the damaged area must be visible to the camera. EyeCat should provide prompts if the user needs to move closer, change the viewing angle, or supply information that an image cannot reveal.
- Has a suitable place to stop. Someone whose bike breaks during a commute needs somewhere safe to inspect it and attempt a repair. EyeCat may need to help the user determine whether their current location is suitable before starting.
- Wants quick, practical help. The user may be under time pressure and needs to understand whether a repair is manageable, which tools it requires, and how long it might take. Tutorials should be optional and available when needed so users can access help without delaying every task.

- Has limited attention to spare. The user may be handling the object while following instructions, so guidance needs to be brief, hands-free, and easy to pause or repeat. EyeCat should present one step at a time and let users ask for more detail.
- Can access the necessary tools and parts. EyeCat provides guidance, but completing a repair depends on having the required equipment. It should identify those requirements before the user begins taking the product apart.
- Is willing to attempt a repair or consider a replacement. The user may want to save money, continue their journey, or keep an existing product working. If repair is impractical, EyeCat should let them move easily to replacement comparisons based on their budget and preferred features.

The biggest assumption is that immediate guidance makes immediate repair possible. Simple setup and accessible tutorials can help both beginners and experienced users get started, but EyeCat still needs to establish whether they have the time, tools, skills, and suitable location to complete the repair. When they do not, it should help them identify another source of assistance or explore replacement options.

How does the interaction flow differ from the other designs?

- **Initiating the task:** With EyeCat, users look at the broken object through their smart glasses and ask for help aloud. The assistant uses camera input and follow-up questions to understand the issue. On a repair website, users must search for the problem and find a suitable guide themselves. With a community repair kiosk, users must travel to the location and present the object or explain the issue.
- **Performing the repair:** EyeCat collects relevant information, presents repair options, and lets users choose their preferred method by voice. Users then follow instructions beside their view of the object while keeping their hands free. They can ask questions, pause, or repeat steps throughout the process. Website users switch their attention between the guide and the object, while kiosk users follow the available guidance and may have access to tools or in-person support.
- **Completing the task:** EyeCat prompts users to test the object and report whether the repair worked. If the issue remains, it can suggest further checks or additional help. Website users may need to search for another guide themselves, while kiosk users may be able to check their repair with assistance before leaving.

How is this design informed by findings from Milestone 1?

Because repair information is spread across many apps, websites, and online guides, we designed EyeCat to bring relevant resources together through one accessible assistant. We chose smart glasses as the main interface so users could

receive help wherever a problem occurs while keeping their hands free to work on the object. Camera input allows users to show EyeCat what they are looking at, reducing the need to know technical terms or describe an unfamiliar part. Voice interaction lets users explain symptoms, answer follow-up questions, and request clarification without typing or navigating menus. Rather than presenting a long list of search results, EyeCat would organize repair methods by the tools required, estimated time, and difficulty, helping users choose an approach that fits their situation. We also chose to display instructions or a tutorial beside the user's view of the object so they could follow each step with less interruption. Controls such as pause, repeat, and next would let users work at their own pace. These choices aim to make repair guidance easier to find and follow without requiring a visit to a fixed location. Since some repairs require specialized tools or professional help, EyeCat would also help users recognize when a repair is beyond their available resources or experience.

Idea 3: Community Repair Scheduler (Kiosk)

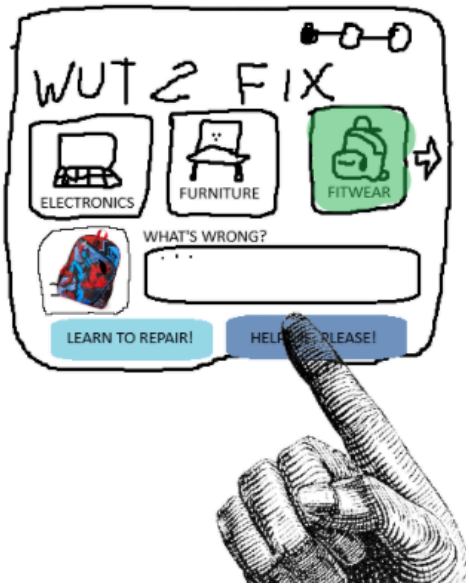
Mockups

Storyboard of Initial Drawings:

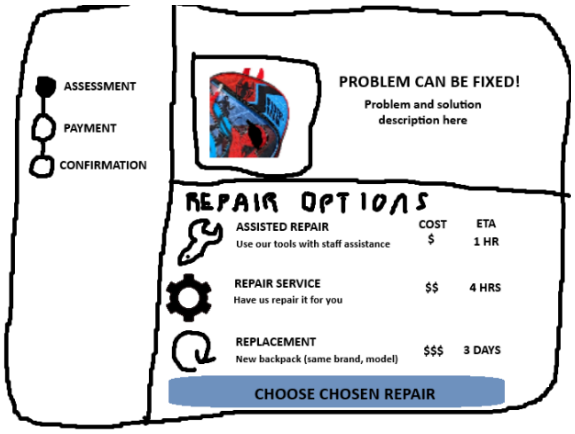
1. Identify Problem



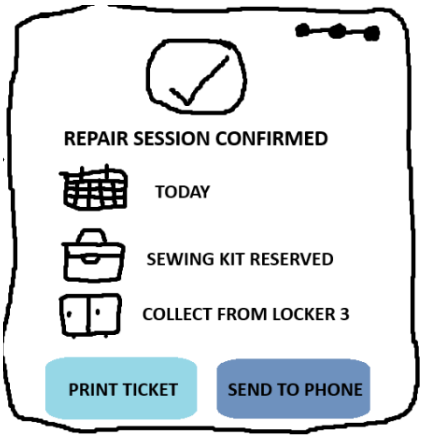
2. Check in at Kiosk



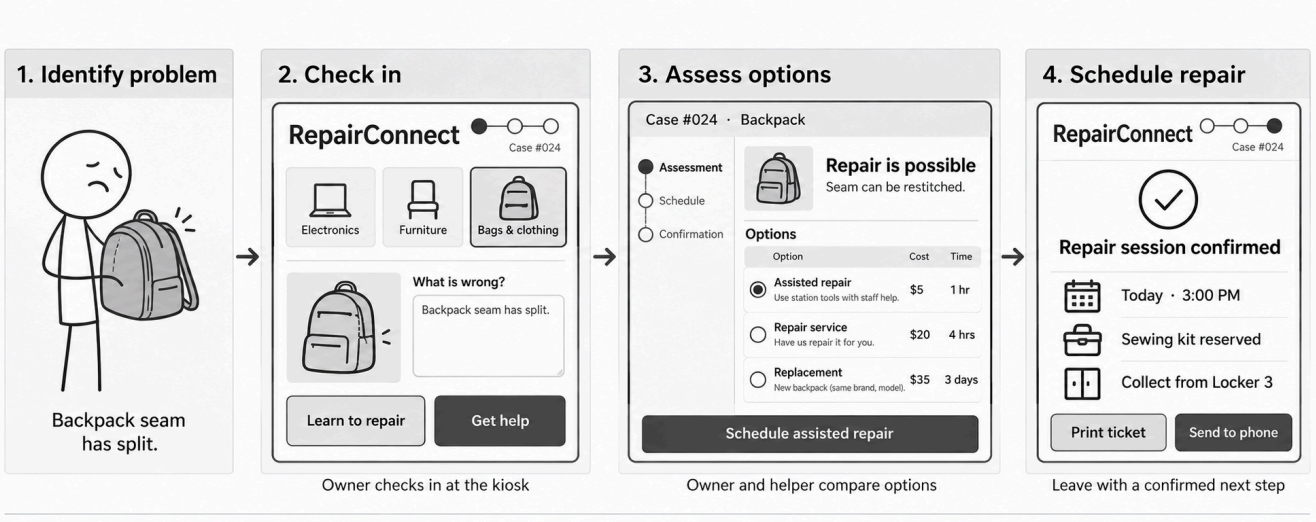
3. Assess available options



4. Schedule your repair



Polished Interface Flow with Task Process Design



Detailed Narrative

The idea would exist as a kiosk in a community space where users could bring a damaged item and decide whether repairing or replacing it makes more sense, based on consumer factors like available budget and time. After selecting the type of item and describing the problem to a volunteer monitoring the station, the volunteer would assess the damage, take a picture of the damaged item for logging purposes, and present different options using the kiosk interface. As shown in the storyboard, an example scenario could be so: someone with a damaged backpack could compare an assisted repair, a professional repair service, and a replacement based on the cost and estimated completion time. If the user chooses an assisted repair, the kiosk could schedule a session, reserve the tools needed, and direct the user to a locker where the repair kit can be collected. The confirmation could then be printed or sent to the user's phone, depending on consumer preference and technological availability. Unlike the website and smart glasses, this idea connects digital guidance with physical tools, staff /volunteer assistance, and a dedicated repair space with reserved tools, help, and time. However, this would require users to travel to a kiosk and depend on the availability of appointments, tools, and staff.

At the conclusion of a repair session, resource return will be handled by the volunteer and the kiosk will be used again to log success in repairs and close the case using a similar interface to the initial workflow. If a desired resource, such as the shown-above sewing kit, or volunteer is unavailable, users will be given the option to either reschedule their appointment to a day where the resource is guaranteed to be available, obtain a referral, or save their case to be opened at a later time.

Design Checklist

What makes this design fundamentally different from the other two?

- The community repair kiosk connects users with a physical location, tools, and staff assistance. The website and smart glasses serve to provide the user with information and virtual guidance on the best way to solve their current dilemma, while the kiosk helps users access the resources needed to complete a repair. It also follows a more structured process in which users compare their options, schedule a repair session, reserve tools, and receive confirmation, enabling further avenues of exploration compared to the traditionally linear style of the website and smart glasses interfaces.

What assumptions does this design make about the user and their situation?

- Can travel to the kiosk with the damaged item.
- Has enough time to attend a scheduled repair session.
- Is comfortable using a touchscreen and following the instructions.
- Wants to compare repairing and replacing the item before making a decision.
- Is willing to attempt a repair or allow someone else to repair the item.
- Has access to a phone or can keep a printed confirmation ticket for extended periods of time.
- Lives near or is willing to drive to a location with available staff, tools, and repair appointments.

Above all, the largest assumption is that the damaged item is portable and that the community repair location has the needed resources available.

How does the interaction flow differ from the other designs?

- **Initiating the task:** The user brings a damaged item to the kiosk, selects its category, and describes the problem to another individual, oftentimes one with adequate expertise. The other two alternatives base analysis off of independent user input with a lower level of item contextualization.
- **Assessing the options:** The kiosk displays and compares options for direct assisted repair, professional repair, and replacement based on cost and completion time. This differs from the other two alternatives through its insource supply of assistance and tools, providing resources directly through the system as opposed to redirects and suggestions of replacement websites or repair services.
- **Completing the task:** the user selects an option, schedules a session, reserves the needed tools, and receives a printed or digital confirmation. All other preparation is handled by the community space in which the kiosk service is located, whereas the other two alternatives directly rely on user preparation, scheduling, and purchasing of tools by themselves.

How is this design informed by findings from Milestone 1?

- Our Milestone 1 research found that cost is one reason people continue using damaged products instead of repairing or replacing them. We also found that access to tools, knowledge, money, and community support to determine whether a repair is possible. The kiosk addresses these issues by placing repair information, cost comparisons, tools, and assistance in one location. It also builds upon the feedback we received about focusing on repair vs replace. Classmates stated that broader money-saving spaces were very oversaturated, this design focuses on providing physical repair resources instead of becoming another budgeting app.

Strengths and Limitations

As touched upon previously, this design alternative excels in providing the consumer with almost everything needed to streamline the process from damage assessment to the end-of-repair workflow. By providing direct means of replacement or access to repair tools, this system's hybrid design greatly benefits its applicability and realistic implementation into a real-world setting. Likewise, the presence of human assistance and intervention provides the strongest accuracy in hands-on damage analysis, recognizable and comfortable conversation and interaction, and damage repair education. However, the introduction of external human and geographical components are also this system's greatest weakness. By introducing these other dependencies, like being dependent on the host's hours and location, which may be inconvenient to some users, staffing availability and inventory costs, queues with multiple cases, unavailability of resources, and failure to accommodate larger items like vehicles, this form of implementation may be incredibly difficult to not only manage, but fund, which is an incredibly important factor in allowing the product to be accessible to as many consumers as possible.

Design Alternative Rationale

In determining what made the most optimal design alternatives that accurately reflected student feedback from milestone 1 and group interest in a problem space, the group sought to better understand the scope and, more importantly, the modality of the designs to open the gateway for new ideas and more thorough approaches to a conceptually simple, yet executionally complicated. As such, the group's three selected modalities of a website, smart glasses-powered voice assistant, and community service kiosk were all carefully selected to capture different, yet complete workflows that help solve the same problem from multiple perspectives based on user ability, financial freedom, and repair feasibility. Furthermore, addressing an existing problem through a different modality not properly recognized on a grand scale gave the group a plethora of options in tackling new aspects of the field that may better address the problem space than existing technologies.

However, as convenient as it would have been, the group's three chosen solutions were not the first of their kind. In brainstorming, the group talked through each example construed, fleshing out as much detail as possible to see if an idea's potential life would last longer than one or two minutes of group discussion before coming to the conclusion it wouldn't work. The group's first, and perhaps most obvious, idea was a simple phone application in which users could type in the name of their damaged item or scan a picture of it, to which the software would then cross search similar or outright identical repair situations across the Internet, using artificial intelligence to determine the best course of action (repair or replace) and give options for each use case. While this would have been the most feasible solution and the easiest to implement, much of our initial research into existing technologies already concerned mobile applications with similar item lookup capabilities, and as our group failed to rationalize how our approach would be addressing a different part of the problem, the idea was ultimately scrapped.

Another scrapped idea the team analyzed were interactive directories/wayfinding kiosks, akin to the devices at the mall that display a map of the mall users can interact with to pinpoint where they want to go. Although the group did not intend on using this technology for wayfinding purposes, the principle of a large screen in a public setting users can interact with remained the driving force in the group's brainstorming discussion. The idea was to place these interactive screens in large common spaces such as malls and equip them with cameras for users to show their damaged products to. The screens would display an idle message like "Got something broken? Let us help you fix it!", upon which interacting with the screen would ask the user to describe their item and its issue before displaying their damaged item or a picture of it, if they didn't have it on them. From there, the machine would analyze the damage and determine potential solutions, pointing to locations within the mall users could visit to purchase

replacements, buy tools, or have their item repaired for them. This design alternative offered a unique way to promote smaller businesses while also offering a convenience to customers, a dynamic somewhat difficult to address in modern business environments. However, this idea was completely hindered by one fatal flaw: item portability and screen dimensions. The design derived itself from the assumption that users would be easily able to access a visualization of their problem, which may not be something they conceptualize until they are aware the technology exists. For instance, if a user has a torn couch, it's unlikely they will have a picture of the damage to show the kiosk, so they will instead think "I wish I had a picture of my couch to show it" and move on with their day. A low barrier of entry and accessibility does not serve much good if strong assumptions of the users are made. Likewise, usage would likely be discouraged by the large, oftentimes clunky screen dimensions, which although great for grabbing user attention from further away, make navigation of the system incredibly tedious when needing to type or press buttons higher up than they are able to comfortably reach. As such, this idea was scrapped in favor of the more accessible, user-assisted community repair kiosk of idea 3.